



SIMATS Online Programming Lab
JAVA PROGRAMMING

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PRACTICE PROGRAMS-06/08/26

← Back

QUESTIONS

Q1
Student
10 marks

Q2
ATM
10 marks

Q3
TAX
10 marks

Q4
BONUS
10 marks

4/4 answered

Q1 Student

10 marks

Write a program to enter the mark of a student in four subjects. Then calculate the total and aggregate. Display the grade obtained by the student, if student scores an aggregate greater than 75% then grade id distinction

Your Code

```
1 public class StudentGrade {
2     public static void main(String[] args) {
3         int m1 = 80;
4         int m2 = 75;
5         int m3 = 90;
6         int m4 = 85;
7         int total = m1 + m2 + m3 + m4;
8         double aggregate = total / 4.0;
9         System.out.println("Total = " + total);
10        System.out.println("Aggregate = " + aggregate);
11        if (aggregate > 75)
12            System.out.println("Grade: Distinction");
13        else
14            System.out.println("Grade: No Distinction");
15    }
16 }
```

Input

stdin input

Output

Total = 330
Aggregate = 82.5
Grade: Distinction

Run

Save



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← Back

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BONUS
10 marks

4/4 answered

Q2 ATM

10 marks

write a program to print the total amount available in the atm machine with the conditions applied total denominations are 2000, 500, 200,100, get the denomination priority from the user and the total number of notes from the user to display the total available balance to the user

Your Code

```
1 public class ATMBalance {
2     public static void main(String[] args) {
3         int denomination = 500;
4         int notes = 20;
5         int total = denomination * notes;
6         System.out.println("Denomination = " + denomination);
7         System.out.println("Number of Notes = " + notes);
8         System.out.println("Total Available Balance = " + total);
9     }
10 }
```

Input

stdin input

Output

Denomination = 500
Number of Notes = 20
Total Available Balance = 10000

Run

Save

PRACTICE PROGRAMS-06/08/26

← Back

QUESTIONS

Q1
Student
10 marksQ2
ATM
10 marksQ3
TAX
10 marksQ4
BONUS
10 marks

4/4 answered

Q3 TAX

10 marks

Write a program to calculate tax given following condition :

If income is less than or equal to 150000 then to tax

If taxable income is 150001 – 300000 the charge 10% tax

If taxable income is 300001 – 500000 the charge 20% tax

If taxable income is above 500001 the charge 30% tax

Your Code

```
1 public class TaxCalculator {
2     public static void main(String[] args) {
3         double income = 350000;
4         double tax;
5         if (income <= 150000)
6             tax = 0;
7         else if (income <= 300000)
8             tax = income * 0.10;
9         else if (income <= 500000)
10            tax = income * 0.20;
11        else
12            tax = income * 0.30;
13        System.out.println("Income = " + income);
14        System.out.println("Tax = " + tax);
15    }
16 }
```

Run

Save

Input

stdin input

Output

```
Income = 350000.0
Tax = 70000.0
```

PRACTICE PROGRAMS-06/08/26

← Back

QUESTIONS

Q1
Student
10 marksQ2
ATM
10 marksQ3
TAX
10 marksQ4
BONUS
10 marks

4/4 answered

Q4 BONUS

10 marks

a bonus 5% salary is given to grade A workers and 10% bonus to Grade B , write a program to enter salary and grade of employee , if salary less than employee gets an extra 2% bonus on salary. Calculate the bonus that has to be given to the employee and print the salary that the employee will get

Your Code

```
1 public class EmployeeBonus {
2     public static void main(String[] args) {
3         double salary = 18000;
4         char grade = 'A';
5         double bonus = 0;
6         if (grade == 'A')
7             bonus = salary * 0.05;
8         else if (grade == 'B')
9             bonus = salary * 0.10;
10        if (salary < 20000)
11            bonus = bonus + (salary * 0.02);
12        double totalSalary = salary + bonus;
13        System.out.println("Salary = " + salary);
14        System.out.println("Grade = " + grade);
15        System.out.println("Bonus = " + bonus);
16        System.out.println("Total Salary = " + totalSalary);
17    }
18 }
```

Run

Save

Input

stdin input

Output

```
Salary = 18000.0
Grade = A
Bonus = 1260.0
Total Salary = 19260.0
```

EXCEPTION-06/08/26

← Back

QUESTIONS

Q1
Handling ArithmeticException
10 marks**Q2**
Handling
ArrayIndexOutOfBoundsException
10 marks**Q3**
Handling
NumberFormatException
10 marks**Q4**
Handling Multiple Exceptions
10 marks

4/4 answered

Q1 Handling ArithmeticException

10 marks

Write a Java program to divide two integers entered by the user. If the denominator is zero, handle the exception using a try-catch block and display an appropriate error message.

Your Code

```
1 import java.util.Scanner;
2 public class ArithmeticExceptionDemo {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         int num = sc.nextInt();
6         int den = sc.nextInt();
7         try {
8             int result = num / den;
9             System.out.println("Result = " + result);
10        } catch (ArithmeticException e) {
11            System.out.println("Error: Division by zero is not allowed");
12        }
13        sc.close();
14    }
15 }
```

Input

10
0

Output

Error: Division by zero is not allowed.

Run

Save

EXCEPTION-06/08/26

← Back

QUESTIONS

Q1
Handling ArithmeticException
10 marks**Q2**
Handling
ArrayIndexOutOfBoundsException
10 marks**Q3**
Handling
NumberFormatException
10 marks**Q4**
Handling Multiple Exceptions
10 marks

4/4 answered

Q2 Handling ArrayIndexOutOfBoundsException

10 marks

Write a Java program to create an integer array of five elements. Ask the user to enter an index and display the corresponding element. If the index is outside the valid range, handle the exception and display an error message.

Your Code

```
1 import java.util.Scanner;
2 public class ArrayIndexOutOfBoundsExceptionDemo {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         int[] arr = {10, 20, 30, 40, 50};
6         int index = sc.nextInt();
7         try {
8             System.out.println("Element: " + arr[index]);
9         } catch (ArrayIndexOutOfBoundsException e) {
10            System.out.println("Error: Invalid array index.");
11        }
12        sc.close();
13    }
14 }
```

Input

10

Output

Error: Invalid array index.

Run

Save

EXCEPTION-06/08/26

← Back

QUESTIONS

Q1
Handling ArithmeticException
10 marks**Q2**
Handling
ArrayIndexOutOfBoundsException
10 marks**Q3**
Handling
NumberFormatException
10 marks**Q4**
Handling Multiple Exceptions
10 marks

4/4 answered

Q3 Handling NumberFormatException

10 marks

Write a Java program to accept a number as a string and convert it into an integer using Integer.parseInt(). If the input is not a valid integer, handle the exception and display an appropriate message.

Your Code

```
1 public class NumberFormatExceptionDemo {
2     public static void main(String[] args) {
3         String str = "WWWWW";
4         try {
5             int num = Integer.parseInt(str);
6             System.out.println("Number: " + num);
7         } catch (NumberFormatException e) {
8             System.out.println("Error: Invalid integer.");
9         }
10    }
11 }
```

Input

stdin input

Output

Error: Invalid integer.

▶ Run

⬇ Save

EXCEPTION-06/08/26

← Back

QUESTIONS

Q1
Handling ArithmeticException
10 marks**Q2**
Handling
ArrayIndexOutOfBoundsException
10 marks**Q3**
Handling
NumberFormatException
10 marks**Q4**
Handling Multiple Exceptions
10 marks

4/4 answered

Q4 Handling Multiple Exceptions

10 marks

Write a Java program that performs the following operations:

Accept two integers from the user.

Divide the first number by the second.

Store the result in an array.

Handle both ArithmeticException and ArrayIndexOutOfBoundsException using multiple catch blocks.

Your Code

```
1 public class MultipleExceptionDemo {
2     public static void main(String[] args) {
3         int a = 10;
4         int b = 2;
5         int[] arr = new int[2];
6         try {
7             if (b == 0)
8                 throw new ArithmeticException();
9             int result = a / b;
10            if (2 >= arr.length)
11                throw new ArrayIndexOutOfBoundsException();
12            arr[2] = result;
13            System.out.println(arr[2]);
14        } catch (ArithmeticException e) {
15            System.out.println("Error: Cannot divide by zero.");
16        } catch (ArrayIndexOutOfBoundsException e) {
17            System.out.println("Error: Array index out of bounds.");
18        }
19    }
20 }
```

Input

stdin input

Output

Error: Array index out of bounds.

▶ Run

⬇ Save